



Evolutionary implications of large-scale patterns in the ecology of Trinidadian guppies, *Poecilia reticulata*

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Previous investigations of the guppy, *Poecilia reticulata*, in Trinidad have demonstrated rapid population differentiation at small geographic scales. However, most studies to date have focused on localities in Trinidad's Northern Range where barrier water falls mark sharp discontinuities in the selection regime and limit scope for gene flow. There is little information on the ecology of the guppy in the rest of the island even though its distribution amongst fish communities and habitats has important evolutionary implications. To determine how large-scale distribution patterns might affect the evolutionary potential of the guppy we surveyed 80 sites representative of a broad range of freshwater environments. We found guppies, which occurred in 80% of our samples, to be the most widely distributed freshwater fish in Trinidad. Guppies are common in predator-rich and turbid habitats, precisely those localities where female preferences are likely to be undermined. Moreover, the widespread distribution of this adaptable species, combined with its promiscuous mating system, may promote gene flow across geographical scales that transcend local selection regimes. These factors are likely to impede the evolution of reproductive isolation.

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ADDITIONAL KEYWORDS: Trinidadian guppy – reproductive isolation – local adaptation – diversity – pollution.

INTRODUCTION

The guppy, *Poecilia reticulata*, is a celebrated case of evolution in action (Endler, 1995). This small fish is native to freshwater streams and rivers in Trinidad, as well as being found in neighbouring Venezuela, Tobago and on other Antillean islands. Its significance as an evolutionary model was first recognized by Caryl and Edna Haskins in the 1940s and 1950s (Haskins & Haskins, 1950) when their pioneering studies highlighted the interplay of natural selection (exerted by predators) and sexual selection (through female choice) in shaping male colour patterns. Almost 50 traits are now known to covary with predation regime (Endler, 1995). Indeed, there are few aspects of behaviour, life-history or morphology that are independent of it (Houde, 1997). The biological significance of Trinidadian guppies is not restricted to this correlation

between the degree of risk and the expression of behavioural and morphological traits; it also arises from the fact that the system has been used to provide an experimental demonstration that natural selection leads to evolution. A series of transplant experiments, in which guppies were translocated from high-risk sites, where they were exposed to many predators, to ones where predation risk was absent or minimal, confirmed that heritable changes in morphology, life-history and behaviour ensue within 10^1 to 10^2 generations (Endler, 1980; Magurran, 1998; Reznick *et al.*, 1997).

One of the great strengths of the guppy system is that investigators have recognized the evolutionary significance of fine-scale variation in ecological factors, such as predation and productivity (Reznick, Butler & Rodd, 2001). Remarkably, given the intense interest in the ecology and evolution of the species, there has been no systematic study of its distribution in Trinidad to date. [The goals of two earlier surveys, by Kenny (1995) and Price (1955), were to document the Trinidadian freshwater fish fauna and to describe zoogeographic patterns.] This broader perspective is all

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the more urgent because most investigations of wild guppy populations have been restricted to a relatively small, and not necessarily representative, portion of Trinidad – the Northern Range. To assess the evolutionary potential (Foster & Endler, 1999) of the species within Trinidad as a whole we need much better information on its distribution amongst fish communities and habitats. For example, female choice may reinforce population differentiation and ultimately lead to the evolution of reproductive isolation (Endler & Houde, 1995; Lande, 1981; Turner, 1999). However, predators can select against female choice (Houde, 1997; Iwasa & Pomiankowski, 1991) and female guppies tend to be less choosy (Godin & Briggs, 1996; Gong, 1997), as well as more vulnerable to sneaky mating attempts (Magurran & Nowak, 1991; Magurran & Seghers, 1994) in risky environments. At present we do not know the extent to which guppy populations outside the Northern Range co-occur with fish predators, though on the basis of Kenny's (1995) work, we expect the distributions to show considerable overlap.

Predation is not the only factor that inhibits female choice. Species, such as guppies, where visual elements play a major role in courtship decisions (Endler, 1991, 1995; Verrell, 1999), are vulnerable to a loss of water clarity. Investigations of cichlids (Seehausen, van Alphen & Witte, 1997) in Lake Victoria, where water quality is now adversely affected by eutrophication, have shown how the reproductive barriers between recently evolved species break down when females cannot view potential mates. A second question, therefore, relates to the prevalence of guppy populations in polluted, and particularly turbid, habitats. Finally, gene flow has the potential to overwhelm local adaptation and limit population differentiation (Foster & Endler, 1999). Gene flow between upstream and downstream populations has already been documented in several Trinidadian rivers (Becher & Magurran, 2000; Shaw *et al.*, 1994); guppy mating behaviour (Hughes *et al.*, 1999; Kelley, Graves & Magurran, 1999; Magurran, 1996) may be a contributory factor. Given the promiscuous mating system of the species, gene flow will be enhanced if populations are contiguous. The wide distribution of guppies across river systems in Trinidad would thus be consistent with high levels of gene flow.

Our aim in this paper, then, is to assess the evolutionary implications of large-scale patterns in the ecology of Trinidadian guppies. In particular we focus on aspects of distributional ecology associated with local adaptation and female choice. Accordingly, we conducted an extensive survey of Trinidadian rivers to determine the spatial distribution of guppies as well as their occurrence in relation to habitat type, water quality and community composition.

METHODS

Eighty sites, representing all the major drainages, biogeographic regions and river types in Trinidad (Fig. 1) were sampled between 1996 and 1998. Heavy equipment and the need for access prevented us from choosing sample sites on a strictly random basis. Nevertheless, by stratifying the sites (Thompson, White & Gowan, 1998), we ensured that our survey was representative of fish communities in Trinidad.

The section of stream sampled at each site was short enough to be fished thoroughly, yet long enough for all species present to be represented in the catch. The sampling protocol also included the major habitat types, for instance pool and riffle, present in the river at that point. An average length of 50 m was used. The methods and gear employed to sample fish depended on several factors: width and depth of stream; substratum type (for example sand, rock, algae or branches); turbidity and water current speed. Electrofishing was found to provide a better representation of fish assemblage structure than other methods such as seining (Dauble & Gray, 1980). In addition, it could be used in areas such as those with dense vegetation, that were poorly sampled by other gears. Electrofishing was therefore employed wherever conditions were suitable (Perrow, Côté & Evans, 1996), particularly in the clear, stony-bottomed streams of the Northern Range. In the deep pools of these streams, electrofishing was supplemented by visual census techniques (Perrow *et al.*, 1996). Seining was used for most of the survey since the majority of the rivers were turbid and not suitable for electrofishing. Large deep rivers were sampled with gillnets and a trammel net. Guppies and other small fish were collected by dip nets. The identities of all fish species collected [as well as those seen but not caught (Perrow *et al.*, 1996)] were recorded for each site.

Except for the largest rivers, fishing was carried out during the day. Where possible, block nets were first deployed at the upstream and downstream ends of the sampling station. In the large rivers, fishing was carried out at night when the use of the gill and trammel nets would be most effective. The nets were set at or just before nightfall, and retrieved 3–4 h later. They were then reset and retrieved at dawn.

Habitat parameters (temperature, stream width, depth and overhead cover) were measured at three points, and averaged to give a mean value per site. Water quality was assessed by measuring oil and grease, suspended solids, total phosphorus, conductivity, dissolved oxygen (DO), pH and hardness. These parameters were measured either *in situ* (conductivity, DO, pH) or in the laboratory (the remainder). *In situ* measurements were made at the same points at which physical measurements were taken. Where

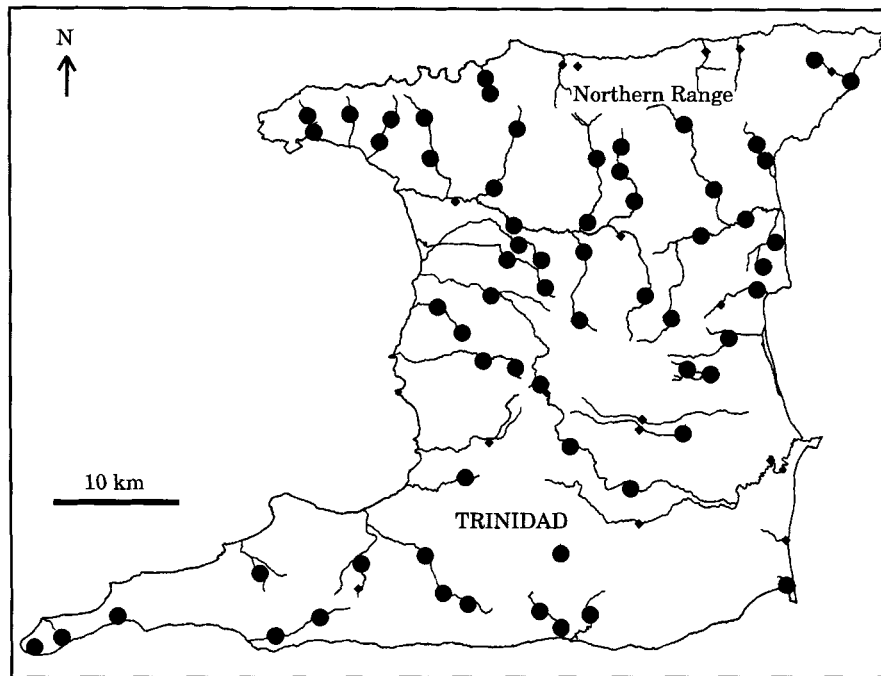


Figure 1. Location of the sample sites in Trinidad indicating where guppies were present (●) or absent (◆).

measurements were to be made in the laboratory, duplicate or triplicate samples were collected at the upstream end of the sampling site. Each collection was made midway between the river-banks, and at mid-depth in the water column. Water for oil and grease determination was collected at the water surface and preserved with nitric acid for later analysis. Other samples were stored on ice and, in most cases, analysed the same day.

Measurements were made with a HACH water quality test kit. Other methods were substituted in cases where water colour and/or turbidity affected the reliability of the results obtained. Where feasible, chemical parameters were measured using USEPA accepted standard methods (American Public Health Association, 1995).

RESULTS

We discovered that guppies are indeed widely distributed across Trinidad (Fig. 1). The main region devoid of the species is the relatively isolated northern fringe of the island. It is also absent from a number of localities in the south-eastern portion of Trinidad. Nonetheless, guppies occur in 80% of sites sampled (Fig. 2) and are, by a considerable margin, the most widespread fish. Three of the four next most common species, *Astyanax bimaculatus*, *Aequidens pulcher*, and *Rivulus hartii*, usually considered minor predators of guppies, are found in 54%, 53% and 48% of sites respectively. *A. pulcher* is classed as an epibenthic

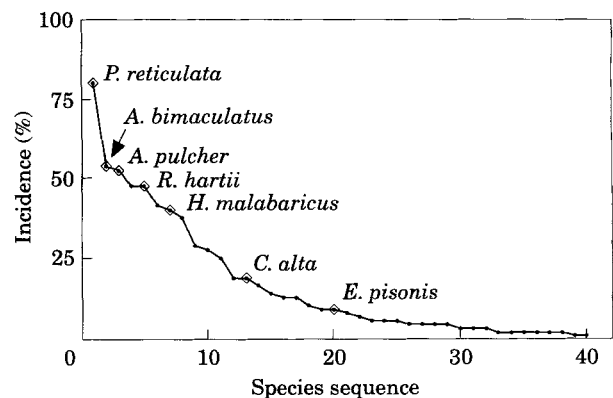


Figure 2. The percentage incidence (that is the number of sites in which a species was found as a percentage of the total number of sites) of the 40 freshwater species recorded in the survey. Species are arranged in sequence from most abundant to least abundant. The identities of 3 major and 3 minor guppy predators (see text) are indicated. Guppies are the most widespread of all species.

invertebrate predator and *A. bimaculatus* as a mid-water invertebrate feeder/omnivore (Winemiller, 1991). *R. hartii* targets immature guppies and presents little threat to adult fish (Seghers, 1973). *Hoplias malabaricus*, categorized by Winemiller (1991) as an ambush fusiform piscivore, is present in 40% of our samples, and emerges as the most widely distributed of the major guppy predators. In contrast, the distribution of *Crenicichla alta*, often viewed as the key

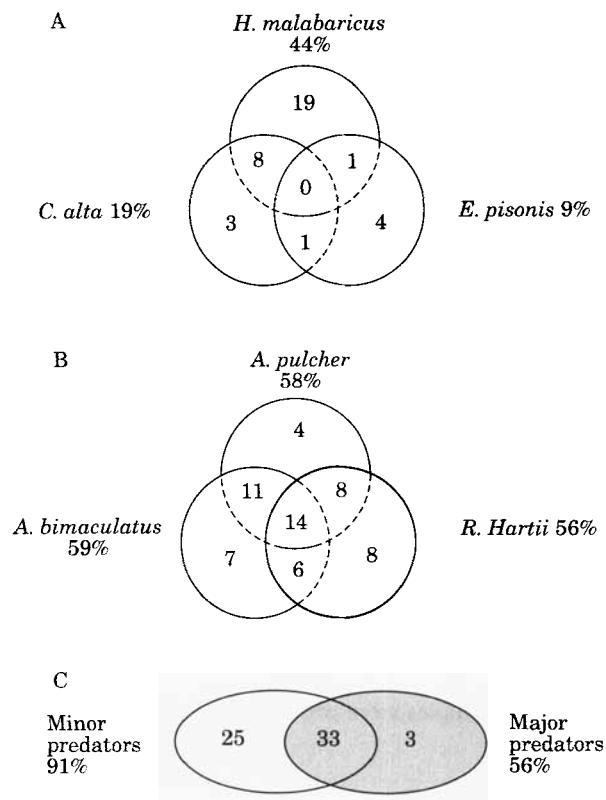


Figure 3. The occurrence of these major and minor predators in guppy populations. Numbers within the venn diagrams are the frequencies of guppy localities at which each combination of predators is found. (A) illustrates the conjoint distributions of three major predators with each other and with guppies. The same analysis is shown for three minor predators in (B) while (C) indicates the overlap between these major and minor predators. Information on potential predators was taken from Houde (1997).

guppy predator (Haskins *et al.*, 1961) is more restricted. It occurs in 19% of localities. *H. malabaricus* and *C. alta* belong to the South American fauna that dominates most of Trinidad. This is replaced by an Antillean fauna on the northern slopes of the Northern Range (Kenny, 1995). Here guppies face a different suite of enemies. Of these, the predatory goby, *Eleotris pisonis* (Reznick & Bryga, 1996), which occurs in 9% of localities overall, is most common.

Guppy predators are clearly abundant in Trinidad. To what extent do the distributions of predators and prey coincide? Figure 3A shows that 56% of guppy populations are exposed to one or more of these three major predators. While *H. malabaricus* is present in twice as many guppy localities as *C. alta*, 50% of populations co-occur with one or both of these piscivores. Minor predators, found in 91% of guppy populations (Fig. 3B), are even more broadly distributed. In total, at least one of these six species is found in 95% of guppy localities (Fig. 3C).

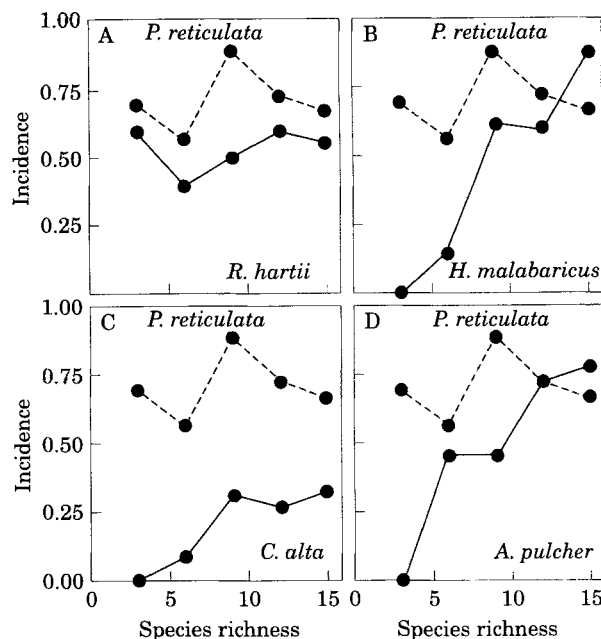


Figure 4. The incidence function (Diamond, 1975) of guppies (dotted line illustrated on all graphs) and *A. R. hartii*, *B. H. malabaricus*, *C. C. alta*, and *D. A. pulcher* in fish communities of different richness levels (1–3 species, 4–6 species, 7–9 species, 10–12 species and 13–15 species). The incidence function plots the proportional incidence of a species in communities of different sizes. There is a significant increase in the incidence of *H. malabaricus* ($F_{1,3}=36.0$, $P=0.01$), *C. alta* ($F_{1,3}=13.7$, $P=0.03$) and *A. pulcher* ($F_{1,3}=21.0$, $P=0.02$) in richer communities, but not of *P. reticulata* ($F_{1,3}=0.06$, $P=0.83$) or *R. hartii* ($F_{1,3}=0.16$, $P=0.72$).

Guppies occur not only in a wide variety of fish communities, but also in assemblages of varying diversity. As Figure 4 reveals, the incidence of guppies, that is the proportion of communities to which they belong (Diamond, 1975), remains constant as the species richness of the assemblages increases from 1 to 15 fish species (the maximum in our study). However, the co-occurrence of guppies with their major predators is a function of assemblage size. *C. alta* and *H. malabaricus* are significantly more prevalent in species rich communities (Fig. 4) (*E. pisonis* was recorded too infrequently for this analysis). *A. pulcher* (Fig. 4) and *A. bimaculatus* (not illustrated but see Phillip, 1998) are also more likely to be found in diverse assemblages. In contrast, although *R. hartii* is less common than the guppy, its distribution is similarly independent of community size.

Are guppies also ubiquitous in terms of the habitats they occupy? Figure 5 compares the localities where guppies are found with non-guppy sites on the basis of three measures – temperature, turbidity (as suspended solids), and biochemical oxygen demand (BOD). Table

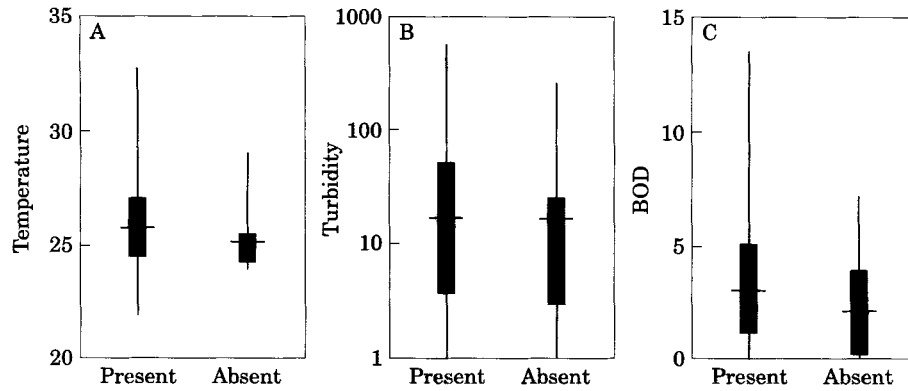


Figure 5. The median, interquartile range and total range of three habitat variables (A) temperature (°C), (B) turbidity (suspended solids – mg.l⁻¹), and (C) BOD (biochemical oxygen demand – mg.l⁻¹) for the $n=64$ sites where guppies were present and the $n=16$ sites from which they were absent.

Table 1. Comparison of ‘guppy sites’ and ‘non-guppy sites’. Median values of each habitat variable and water quality parameter are listed. Mann–Whitney U tests compare the sites; z and P^1 values are given

Habitat variable	Guppy sites ($n=64$) Median	Non-guppy sites ($n=16$) Median	z	P	
Temperature (°C)	25.8	25.2	-0.87	0.38	ns
Width (m)	3.8	4.7	-1.09	0.28	ns
Depth (m)	0.42	0.54	-1.77	0.08	ns
Forest cover (%)	43.3	56.6	-1.71	0.24	ns
pH	7.2	7.2	-0.86	0.39	ns
Conductivity (µs)	323	233	-0.75	0.45	ns
Hardness (mg.l ⁻¹)	108	69	-1.28	0.20	ns
Total suspended solids (mg.l ⁻¹)	17	17	-0.53	0.60	ns
Dissolved oxygen (mg.l ⁻¹)	5.4	5.7	-0.35	0.73	ns
BOD (mg.l ⁻¹)	2.9	2.0	-2.09	0.03	*
P (mg.l ⁻¹)	0.18	0.05	-2.14	0.03	*
Oil and grease (mg.l ⁻¹)	3.5	0.0	-2.60	0.006	**

¹Strictly speaking, because these are multiple comparisons on the same data set, a ‘Bonferroni’ correction should be applied. None of the observed P values fall below this conservative threshold, but the conclusion, that guppies are prevalent in sites with poor water quality, is unaffected.

1 lists the median values of these and the other biotic and abiotic parameters. Given the fact that there are more guppy sites ($n=64$) than non-guppy ones ($n=16$) it is perhaps not surprising that the range of values is greater in the former case (Fig. 5). With one notable exception, however, there are no significant differences between the two types of site (Table 1). The exception relates to the water quality parameters that are associated with pollution: BOD, phosphorus and oil and grease. In each case the guppy sites are more polluted. This is in part a consequence of the fact that guppies tend to occur in all but the most remote sites (which also happen to be the least polluted ones; Phillip, 1998). However, it also reveals that guppies can tolerate poor water quality and that they are not restricted to

oligotrophic mountain streams. There is a positive relationship between assemblage richness (indicative of the presence of potential predators) and turbidity ($r_s=0.37$, $z=3.89$, $P<0.0001$).

DISCUSSION

The versatility of the guppy, in terms of the communities and habitats it occupies, is clearly illustrated by the results. The species is the most widely distributed fish in Trinidad, present in all river systems apart from some northerly draining streams. It is conceivable that guppies are absent from these Antillean communities because they are in some way unsuitable. However, a number of Trinidadian rivers,

such as the Yarra, Marianne and Paria, that fall within the Antillean zoogeographic region (Kenny, 1995), support thriving populations of the species (Rodd & Reznick, 1991). The most likely explanation for the sparse distribution of guppies in rivers to the north of the Northern Range (see also Kenny, 1995) is that the remoteness of these sites has prevented successful colonization. In contrast, the absence of guppies from some of the large rivers in Central and Southern Trinidad is probably an artefact of sampling. These were the sites that were sampled at night, and thus the ones in which small fish were most likely to be overlooked. However, this does not detract from the conclusion that guppies are broadly distributed. They are found in both turbid and clear water sites, in enriched habitats as well as oligotrophic ones, and are one of the last species to persist in heavily polluted rivers (Phillip, 1998). In short, there are few freshwater habitats accessible to guppies that have not been successfully colonized by them.

It is possible that humans have played a significant role in shaping the contemporary distribution of guppies in Trinidad. After all, the fish are small and easily transported, and have sometimes been employed for mosquito control (Courtenay & Meffe, 1989). There are two arguments against this. First, guppies in different Trinidadian drainages, most notably the Caroni and Oropuche, exhibit marked genetic divergence (Carvalho *et al.*, 1991; Fajen & Breden, 1992). Repeated artificial movements of fish would have disrupted these natural patterns. Second, guppies can quickly and successfully colonize downstream sites (Haskins *et al.*, 1961), as well as upstream ones (barrier waterfalls or intermittent streams typically mark the upper limits of distribution in a river system) and we know from our study that a broad range of freshwater habitats are suitable. It seems improbable that the wide distribution of the guppy in Trinidad is either a recent phenomenon or one that has been significantly modified by human influence.

What then are the evolutionary implications of these distribution patterns? The widespread occurrence of the species in lowland habitats shows that guppies are common in places where female choice may be compromised, either by the presence of predators or by water turbidity. *H. malabaricus* emerged as the predator mostly widely co-incident with guppies. Although the role of *C. alta* in shaping guppy behaviour and evolution is well-documented, other piscivorous fish may also be effective agents of natural selection. Reznick & Bryga (1996), for example, found that the life-histories of guppies in streams on the northern slopes of the Northern Range parallel those in the southern ones even though the predators in these northern sites belong to the Antillean fauna. Given its widespread distribution in Trinidad, *H. malabaricus*

may prove to be even more important than *C. alta* in guppy evolution, though the details of the interactions between the two species remain to be resolved. What is clear from Figures 3 and 4 is that the potential of predation in undermining female choice is considerable.

Turbidity is likely to reinforce the effects of predators on female choice. Neotropical rivers are often naturally sediment laden for all or part of the year (Lowe-McConnell, 1987), and it is probable that, over their evolutionary history in Trinidad and the surrounding region, many lowland guppy populations have experienced turbid conditions. Luyten & Liley (1991) conducted a series of experiments in which they allowed males from two populations to compete for matings. One experiment compared the mating success of males from an upstream locality, the Naranjo, a clear water site, and males from the Guayamare River, a persistently turbid stream. When the trials were conducted in clear water, Naranjo males were significantly more successful than Guaymare males in mating with Naranjo females. If, by contrast, the matings took place in turbid water, the relative success of Naranjo males dropped from 72% (measured by the number of sperm transferred to females) to 58%. Luyten and Liley speculate that the greater dependence of Guayamare males on gonopodial thrusting (sneaky mating attempts) may have accounted for their increased success with Naranjo females in turbid water. The importance of the visual elements of courtship, particularly male colour patterns (Endler & Houde, 1995), but also the sigmoid display (Farr, 1980; Matthews, Evans & Magurran, 1997), will be reduced in circumstances where the female cannot clearly view the male. As a result, females will be less able to exercise choice, the intensity of which will in any case be reduced because turbid sites also tend to be predator rich ones.

Finally, the distribution of the species along the entire length of rivers will facilitate gene flow because individuals that migrate even short distances will find conditions that are suitable and conspecifics with whom they may reproduce. Investigations that have focused on the distribution of guppies in single rivers [the Guanapo River (Gilliam, Fraser & Alkins-Koo, 1993) and the Aripo and Tacarigua Rivers (Shaw *et al.*, 1994)] confirm their occurrence in all but the lowest order streams. Shaw *et al.* (1992) used allozymes and Becher & Magurran (2000) mtDNA to demonstrate that fish that had migrated from the Upper to the Lower Turure River successfully interbred with the indigenous guppies there. Preferences by females for unusual males (Farr, 1977; Hughes *et al.*, 1999) combined with preferences by males for unfamiliar females (Kelley *et al.*, 1999) will contribute to gene flow. Mating

success is also a function of male dominance (Kodric-Brown, 1992), which varies amongst populations (Magurran *et al.*, 1996). If males migrating into a population were more competitive than the native males, gene flow would again be enhanced. The mating system of the guppy, in conjunction with the widespread distribution of this adaptable species, appears to play a key role in impeding the evolution of reproductive isolation.

Interestingly, guppy biologists have hitherto focused their efforts on the upper reaches of rivers where the water is almost always clear and where barrier waterfalls mark sharp discontinuities between predation regimes, as well as inhibiting upstream gene flow. Contrasts in female choice (Houde, 1997) and reproductive strategies (Evans & Magurran, 1999) are especially pronounced in such localities. Our results suggest that population differentiation is unlikely to be as marked in other parts of the guppy's range in Trinidad. The small spatial scale over which this differentiation is evident may also, in part at least, be a product of the special characteristics of Trinidad's Northern Range. It would be informative to measure gene flow and the scale of population differentiation in guppy populations in lowland drainages.

Intuitively, it might seem that species with large range sizes are more likely to diverge (through allopatric or parapatric speciation) than those with small ones. Darwin (1859) was first to make the prediction and, as Gaston & Chown (1999) note, the idea continues to attract support [see for example Tokeshi (1999) and Rosenzweig (1995)]. This is because larger ranges appear to offer more opportunities for fragmentation or subdivision by a barrier, thus facilitating speciation. However, it has recently been argued (Gaston & Chown, 1999) that it is in fact the species with small to intermediate range sizes that are more likely to speciate. Widely distributed species may have good dispersal abilities (Mayr, 1963) which enhance gene flow and promote genetic cohesion (Rice & Hostert, 1993), whereas species with poor dispersal abilities will tend to form patchy populations and thus have higher speciation rates (Gaston & Chown, 1999). While the range of the guppy is not large in a global sense – though it has been massively extended in the last century through human influence (Meffe & Snelson, 1989) – the species is a good disperser and, as we have demonstrated in this paper, is widely distributed across habitats and ecological communities.

In conclusion, it is illuminating to consider large-scale patterns in the distribution of the guppy, not only to set in context the many excellent fine-scale studies that reveal rapid evolution at a population level, but also to make predictions about the evolution of reproductive isolation amongst populations. Although the guppy is an exceptionally well-studied

species this broader perspective has, until now, been largely ignored. The same is undoubtedly true for other species, particularly those that have not acquired the status of model organisms. Considerable emphasis is now being placed on understanding the evolution of biological diversity, and new approaches, such as the analysis of geographical range sizes mentioned above, appear promising.

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